Township of NIPISSING

"Life the way it should be" ESTABLISHED 1888

TOWNSHIP OF NIPISSING

STAFF REPORT

FROM: Dan MacInnis

DATE: March 28th, 2024

Bear Creek Culvert Failure

On March 12th 2024 a sink hole was discovered on Bear Creek Road. Sink hole was filled. Again on March 28th 2024, after a rain storm a sink hole redeveloped at the big culvert at the entrance to Bear Creek Road. The culvert size is 3.2 meters high by 5.5 meters wide x 29 meters long. (10.5 feet x 18 feet wide x 100 feet long.)

After the first failure, I called a culvert relining company to evaluate the existing pipe to see if it could be relined. Relining is an option as the pipe is still holding its original shape. If we choose to hold off on a reline, it may eventually not be an option as the pipe would become too distorted in shape. The cost will increase exponentially if we have to excavate the culvert. We would have to build detours for local traffic to enter and exit as there is only one way in and out.

The Township has 2 options;

Option 1 is a galvanized liner that would be much like what's in the creek now. The cost would be $94,000 for the liner plus labour. (life expectancy is 50 years)

Option 2 is a polymer coated pipe liner that will give the pipe a much longer life expectancy. The cost would be $133,000 for the liner plus labour. (life expectancy is 75 years).
These prices are just to purchase the liner only.

In order to save on cost, The Township will purchase the liner directly from supplier for a cost savings of approximately $2,500.

Estimated costs to install either liner would be $65,000.

My recommendation would be to go with Option 2: The polymer coated liner.

Dan MacInnis, Operations Superintendent